

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (Backlog) Examination (CL) March- April 2018

CL241 Geology for Civil Engineers

Date: 02.04.2018 (Monday)

Time: 01.30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Sections I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1. Answer the questions.

[07]

- (i) The nature of the asthenosphere is _____.
(a) semi liquid (c) gaseous
(b) rigid (d) plastic
- (ii) Moho discontinuity is at the (average) depth of about _____.
(a) 290km (c) 35km
(b) 250km (d) 6000km
- (iii) Which one of following is responsible for the formation of kaolin clay?
(a) Oxidation (c) Thermal expansion
(b) Carbonation (d) Hydration
- (iv) The _____ are the fastest among all the seismic waves.
(a) P waves (c) S waves
(b) Love waves (d) Rayleigh waves
- (v) Granite is a _____ type of rock.
(a) igneous (c) sedimentary
(b) metamorphic (d) metasedimentary
- (vi) Rift valley is the surficial feature associated with _____ boundaries.
(a) transform plate (c) divergent plate
(b) convergent plate (d) None of the above
- (vii) Karst topography is due to action of _____.
(a) water (c) glaciers
(b) wind (d) ground water

Q-2. Answer any three questions. Each carries five marks. [15]

- (i) Explain Crater and cone type of volcanic eruption.
- (ii) Write a note on geological action of groundwater and explain Karst Topography.
- (iii) Explain various sedimentary rocks based on their origin. Cite suitable examples.
- (iv) Write short notes on (a) sediment transportation due to water (b) wind deposition.
- (v) Explain major elements of visual interpretation in Satellite images.

Q-3. Answer the questions. [13]

- (i) How does magnitude and intensity scale differs? Define isoseismal lines. [04]
- (ii) Write cleavage and fracture properties of mineral. [04]
- (iii) Differentiate: (a) Granite and Basalt (b) limestone and sandstone, based on their origin, composition, specific identifiable properties. [05]

OR

- (iii) Write a note on (a) Thermal metamorphism (b) Concordant igneous body [05]

SECTION – II**Q-4. Answer the questions. [07]**

- (i) Match the following terminologies on the left hand side with correct pairs on the right hand side

A. Enechelon fault	i. Hanging wall moved relatively down	[04]
B. Radial fault	ii. Hanging wall moved relatively up	
C. Reversed fault	iii. Fault slip is parallel to the direction of faulting	
D. Normal Fault	iv. Overlapping faults	
	v. Fault emerging from a common center	

- (ii) What will be the strike of sandstone bed having true dip of $S52^{\circ}E$? [03]
 - (a) Towards North
 - (b) Towards East
 - (c) Towards West
 - (d) None of the options
- (iii) The volume of water can be drawn from a given reservoir in a certain interval of time is _____.
 - (a) Storage capacity
 - (b) Reservoir yield
 - (c) Pool level
 - (d) Reservoir height
- (iv) Which of the following is a component of Geoinformatics ?
 - (a) GIS
 - (b) GPS
 - (c) Remote sensing
 - (d) All

Q-5. Answer any three questions. Each carries five marks. [15]

- (i) Draw a neat sketch of Anticline fold and explain the parts.
- (ii) Differentiate between Joint and Fault.
- (iii) Explain Angular unconformity with the help of a net sketch.
- (iv) Explain how rate of weathering, topographic and physiographic conditions govern the site selection for a Dam.
- (v) Explain the source of errors of GPS data.

Q-6. Answer the questions. [13]

- (i) Explain how ground water level governs the selection of reservoir site. [04]
- (ii) Give an outline of Geology of Gujarat. [04]

OR

- (ii) Write a note on components and use of geological maps in civil engineering? [04]
- (iii) Describe the geological investigations that should be undertaken in the different stages of construction of a tunnel. [05]

OR

- (iii) Draw a neat sketch of ANY TWO of the following. [05]
 - a) Recumbent fold
 - b) Reverse fault
 - c) Disconformity
 - d) Horst and Graben
